

TIME SERIES ANALYSIS & CLASSIFICATION

T. MEDKOUR & H. BRAIRI

NATIONAL SCHOOL OF ARTIFICIAL INTELLIGENCE

Chapter X: Introduction to Spectral Analysis

1. Introduction
2. The Periodogram
3. Spectral Density of Time Series
4. Linear Filters
5. Estimating the Spectral Density

INTRODUCTION

Idea

- ▶ Decompose a stationary time series $\{X_t\}$ into a combination of sinusoids, with random (and uncorrelated) coefficients.
- ▶ This is referred to as **spectral analysis** or analysis in the **frequency domain**, in contrast to the time domain approach we have considered so far.
- ▶ The frequency domain approach considers regression on sinusoids; the time domain approach considers regression on past values of the time series.
- ▶ Historically, spectral analysis began with the search for "hidden periodicities" in time series data.

Recall

- ▶ The cosine curve has the equation:

$$x(t) = R \cos(2\pi ft + \Phi)$$

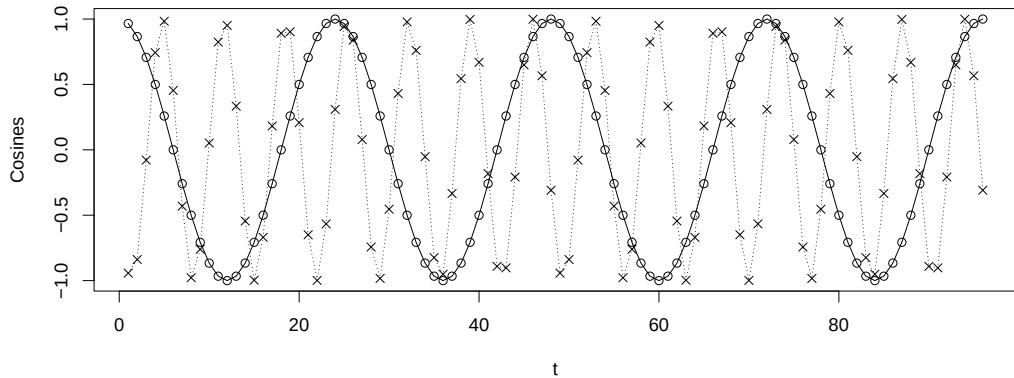
where $R(> 0)$ is the amplitude, f the frequency, and Φ the phase of the curve.

- ▶ The period of the cosine wave is $1/f$ time units, as the curve repeats itself exactly every $1/f$ time units.

Example

- ▶ We consider two discrete-time cosine curves with time running from 1 to 96. The frequencies are $\frac{4}{96}$ and $\frac{14}{96}$, respectively.
- ▶ The lower-frequency curve has a phase of zero, but the higher-frequency curve is shifted by a phase of 0.6π .

Example

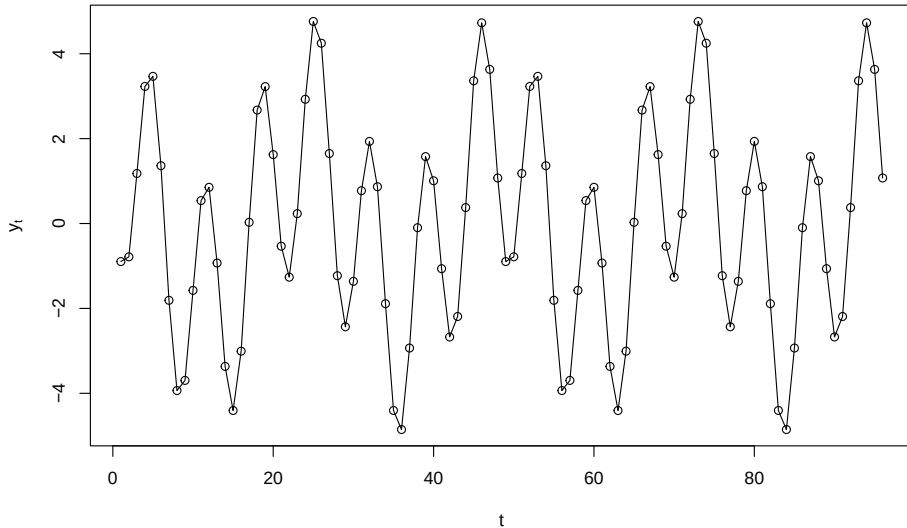


Example

Linear combination of the two cosine curves with a multiplier of 2 on the low-frequency curve and a multiplier of 3 on the higher-frequency curve and a phase of 0.6π

$$Y(t) = 2 \cos \left(2\pi \frac{4t}{96} \right) + 3 \cos \left(2\pi \left[\frac{14t}{96} + 0.3 \right] \right)$$

Example



Reparameterising the Cosine Curve

- ▶ This expression is not convenient for estimation because the parameters R and Φ do not enter the expression linearly. Instead, we use a trigonometric identity to parameterize the equation as:

$$R \cos(2\pi ft + \Phi) = A \cos(2\pi ft) + B \sin(2\pi ft)$$

where:

$$\begin{aligned} R &= \sqrt{A^2 + B^2} & \Phi &= \arctan(-B/A) \\ A &= R \cos(\Phi) & B &= -R \sin(\Phi) \end{aligned}$$

Linear Combination of Cosine Curves

- ▶ A general linear combination of m cosine curves with different amplitudes, frequencies, and phases can be written as:

$$Y(t) = A_0 + \sum_{j=1}^m [A_j \cos(2\pi f_j t) + B_j \sin(2\pi f_j t)]$$

- ▶ If the frequencies f_j are the **Fourier frequencies**, that is of the form j/n , for $j = 1, \dots, n$, the cosine and sine predictor variables are orthogonal

Linear Combination of Cosine Curves

In this case, the least squares estimates are given by:

$$\hat{A}_0 = \bar{Y}$$

$$\hat{A}_j = \frac{2}{n} \sum_{t=1}^n Y_t \cos(2\pi tj/n)$$

$$\hat{B}_j = \frac{2}{n} \sum_{t=1}^n Y_t \sin(2\pi tj/n)$$

Example Fitting

- ▶ If we apply these formulas to the series shown in the example, we would obtain perfect results.
- ▶ At frequency $f_4 = 4/96$, we would obtain $\hat{A}_4 = 2$ and $\hat{B}_4 = 0$, and at frequency $f_{14} = 14/96$, we would obtain $\hat{A}_{14} = -0.927051$ and $\hat{B}_{14} = -2.85317$.
- ▶ We would obtain estimates of zero for the regression coefficients at all other frequencies.

Note

- ▶ Any series of any length n , whether deterministic or stochastic and with or without any true periodicities, can be fit perfectly by the combination of cosines model

$$Y(t) = A_0 + \sum_{j=1}^m [A_j \cos(2\pi f_j t) + B_j \sin(2\pi f_j t)]$$

by choosing $m = n/2$ if n is even and $m = (n - 1)/2$ if n is odd.

- ▶ There are then n parameters to adjust (estimate), in order to fit the series of length n .

THE PERIODOGRAM

The Periodogram

- ▶ For odd sample sizes with $n = 2k + 1$, the periodogram I at frequency $f = j/n$ for $j = 1, 2, \dots, k$ is defined as:

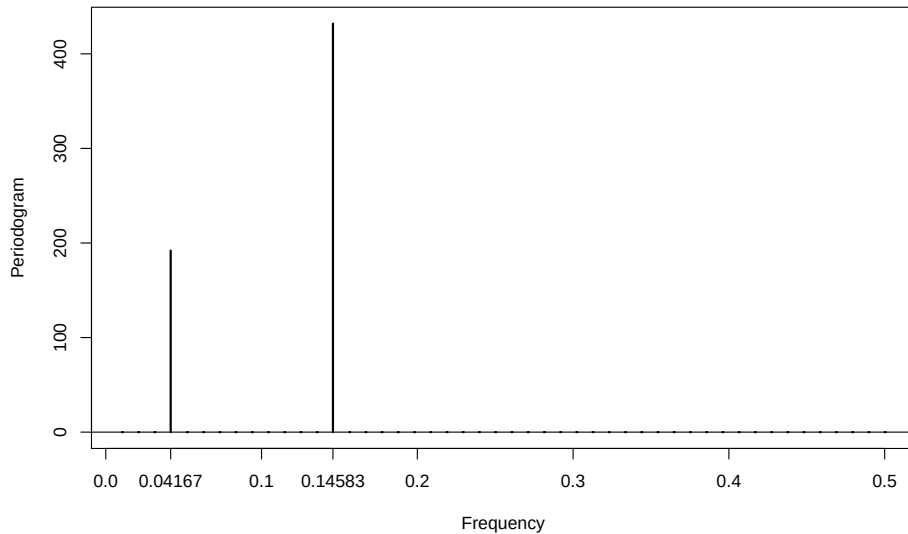
$$I\left(\frac{j}{n}\right) = \frac{n}{2} \left(\hat{A}_j^2 + \hat{B}_j^2 \right)$$

- ▶ If the sample size is even and $n = 2k$, the periodogram is given by:

$$I\left(\frac{j}{n}\right) = \frac{n}{2} \left(\hat{A}_j^2 + \hat{B}_j^2 \right) \quad \text{for } j = 1, 2, \dots, k-1$$

$$I\left(\frac{1}{2}\right) = n \left(\hat{A}_k \right)^2$$

The Periodogram



The Periodogram

- ▶ For our example, the heights show the presence and relative strengths of the two cosine-sine components quite clearly.
- ▶ Note also that the frequencies $4/96 \approx 0.04167$ and $14/96 \approx 0.14583$ have been marked on the frequency axis.

Interpretation of the Periodogram

- ▶ **Relative Strength of Frequencies:** The height of the periodogram shows the relative strength of cosine-sine pairs at various frequencies in the overall behavior of the series.
- ▶ **Analysis of Variance:** The periodogram $I(j/n)$ is the sum of squares with two degrees of freedom associated with the coefficient pair (A_j, B_j) at frequency j/n , so we have:

$$\sum_{j=1}^n (Y_j - \bar{Y})^2 = \sum_{j=1}^k I\left(\frac{j}{n}\right)$$

when $n = 2k + 1$ is odd. A similar result holds when n is even.

Efficient Computation

- ▶ For long series, the computation of a large number of regression coefficients might be intensive.
- ▶ Fortunately, quick, efficient numerical methods based on the fast Fourier transform (FFT) have been developed that make the computations feasible for very long time series.

Periodogram with Noisy Data

- ▶ Does the periodogram work just as well when we do not know where or even if there are cosines in the series? What if the series contains additional "noise"?

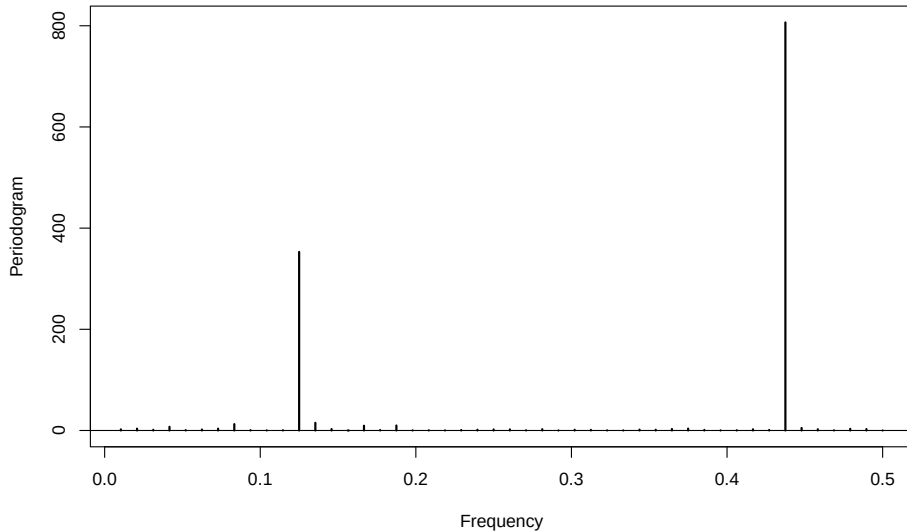
Simulated Example

To illustrate, we generate a time series using randomness to select the frequencies, amplitudes, and phases, and with additional additive white noise. The key elements are:

- ▶ The two frequencies are randomly chosen without replacement from among $1/96, 2/96, \dots, 47/96$.
- ▶ The A 's and B 's are selected independently from normal distributions with means of zero and standard deviations of 2 for the first component and 3 for the second.
- ▶ A normal white noise series, $\{W_t\}$, with zero mean and standard deviation 1, is chosen independently of the A 's and B 's and added on.

$$Y_t = A_1 \cos(2\pi f_1 t) + B_1 \sin(2\pi f_1 t) + A_2 \cos(2\pi f_2 t) + B_2 \sin(2\pi f_2 t) + W_t$$

Simulated Example



Simulated Example

- ▶ The periodogram clearly shows that the series contains two cosine-sine pairs at frequencies of about $12/96 = 0.125$ and $42/96 = 0.4375$ and that the higher-frequency component is much stronger.

Extending the Periodogram

- ▶ Although the Fourier frequencies are special, we extend the definition of the periodogram to all frequencies in the interval $[0, \frac{1}{2}]$:

$$I(f) = \frac{n}{2}(\hat{A}_f^2 + \hat{B}_f^2)$$

$$\hat{A}_f = \frac{2}{n} \sum_{t=1}^n Y_t \cos(2\pi tf)$$

$$\hat{B}_f = \frac{2}{n} \sum_{t=1}^n Y_t \sin(2\pi tf)$$

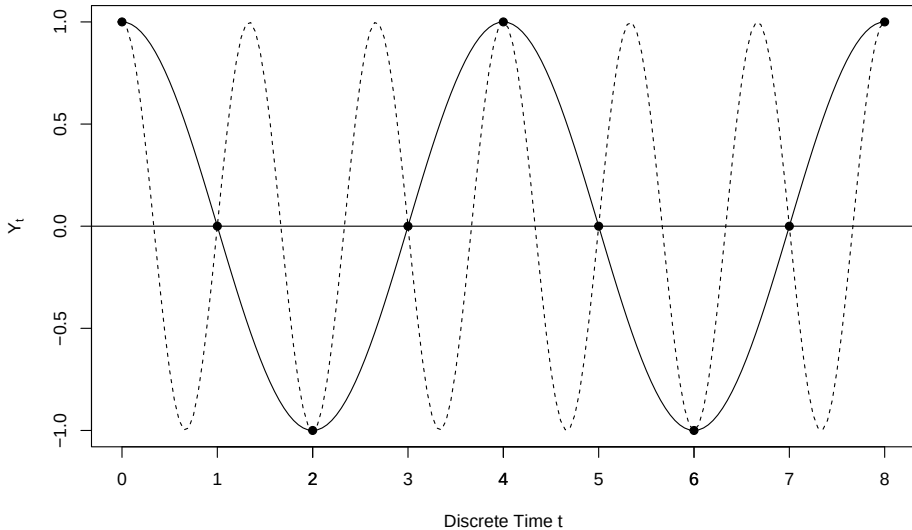
Frequency Interval

- ▶ We restrict frequencies to the interval from 0 to $\frac{1}{2}$ because with discrete-time observations, we could never distinguish between frequencies f and $f + k(\frac{1}{2})$ for any positive integer k .
- ▶ These frequencies are said to be aliased with one another, and it suffices to limit attention to frequencies within the interval from 0 to $\frac{1}{2}$.

Frequency Interval

- ▶ Here we have plotted two cosine curves, one with frequency $f = \frac{1}{4}$ and the other at frequency $f = \frac{3}{4}$.
- ▶ If we only observe the series at the discrete-time points $0, 1, 2, 3, \dots$ the two series are identical. With discrete-time observations, we could never distinguish between these two curves. We say that the two frequencies $\frac{1}{4}$ and $\frac{3}{4}$ are aliased with one another.

Frequency Interval



Spectral Representation

- Consider a time series represented as:

$$Y_t = \sum_{j=1}^m [A_j \cos(2\pi f_j t) + B_j \sin(2\pi f_j t)]$$

where the frequencies $0 < f_1 < f_2 < \cdots < f_m < \frac{1}{2}$ are fixed, and A_j and B_j are independent normal random variables with zero means and $\text{Var}(A_j) = \text{Var}(B_j) = \sigma_j^2$.

Covariance Structure

- ▶ Then the time series $\{Y_t\}$ is stationary with mean zero and:

$$\gamma_k = \sum_{j=1}^m \sigma_j^2 \cos(2\pi k f_j)$$

- ▶ Thus, we can represent γ_k using a Fourier series. The coefficients are the variances of the sinusoidal components.
- ▶ In particular, the process variance γ_0 is a sum of the variances due to each component at the various fixed frequencies:

$$\gamma_0 = \sum_{j=1}^m \sigma_j^2$$

SPECTRAL DENSITY OF TIME SERIES

Spectral Density of a Time Series

- ▶ If a time series $\{X_t\}$ has autocovariance $\gamma(h)$ satisfying

$$\sum_{h=-\infty}^{\infty} |\gamma(h)| < \infty$$

- ▶ Then we define its spectral density as

$$S(\nu) = \sum_{h=-\infty}^{\infty} \gamma(h) e^{-2\pi i \nu h}, \quad \text{for } -\infty < \nu < \infty.$$

Spectral Density: Some Facts

1. $\sum_{h=-\infty}^{\infty} |\gamma(h) e^{-2\pi i \nu h}| < \infty$.
2. S is periodic, with period 1. Thus, we can restrict the domain of S to $-1/2 \leq \nu \leq 1/2$.
3. S is even (that is, $S(\nu) = S(-\nu)$).
4. $S(\nu) \geq 0$.
5. $\gamma(h) = \int_{-1/2}^{1/2} e^{2\pi i \nu h} S(\nu) d\nu$. The autocovariance function and the spectral density are a **Fourier pair**

Example: White Noise

- ▶ For white noise $\{W_t\}$, we have seen that $\gamma(0) = \sigma_w^2$ and $\gamma(h) = 0$ for $h \neq 0$. Thus,

$$\begin{aligned} S(\nu) &= \sum_{h=-\infty}^{\infty} \gamma(h) e^{-2\pi i \nu h} \\ &= \gamma(0) = \sigma_w^2. \end{aligned}$$

- ▶ That is, the spectral density is constant across all frequencies: each frequency in the spectrum contributes equally to the variance.
- ▶ This is the origin of the name white noise: it is like white light, which is a uniform mixture of all frequencies in the visible spectrum.

Example: $AR(1)$

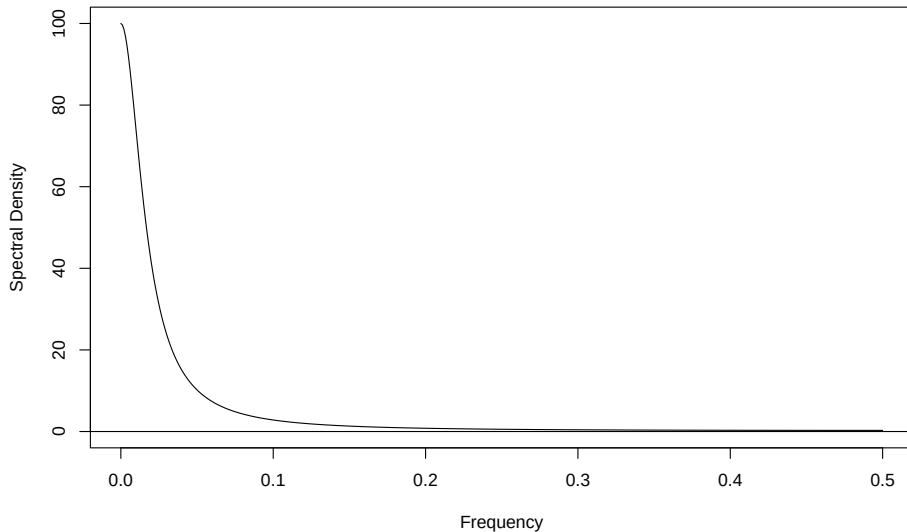
For $X_t = \phi_1 X_{t-1} + W_t$, we have seen that $\gamma(h) = \sigma_w^2 \phi_1^{|h|} / (1 - \phi_1^2)$. Thus,

$$\begin{aligned} S(\nu) &= \sum_{h=-\infty}^{\infty} \gamma(h) e^{-2\pi i \nu h} = \frac{\sigma_w^2}{1 - \phi_1^2} \sum_{h=-\infty}^{\infty} \phi_1^{|h|} e^{-2\pi i \nu h} \\ &= \frac{\sigma_w^2}{1 - \phi_1^2} \left(1 + \sum_{h=1}^{\infty} \phi_1^h (e^{-2\pi i \nu h} + e^{2\pi i \nu h}) \right) \\ &= \frac{\sigma_w^2}{1 - \phi_1^2} \left(1 + \frac{\phi_1 e^{-2\pi i \nu}}{1 - \phi_1 e^{-2\pi i \nu}} + \frac{\phi_1 e^{2\pi i \nu}}{1 - \phi_1 e^{2\pi i \nu}} \right) \\ &= \frac{\sigma_w^2}{(1 - \phi_1^2)} \frac{1 - \phi_1 e^{-2\pi i \nu} \phi_1 e^{2\pi i \nu}}{(1 - \phi_1 e^{-2\pi i \nu})(1 - \phi_1 e^{2\pi i \nu})} \\ &= \frac{\sigma_w^2}{1 - 2\phi_1 \cos(2\pi \nu) + \phi_1^2} \end{aligned}$$

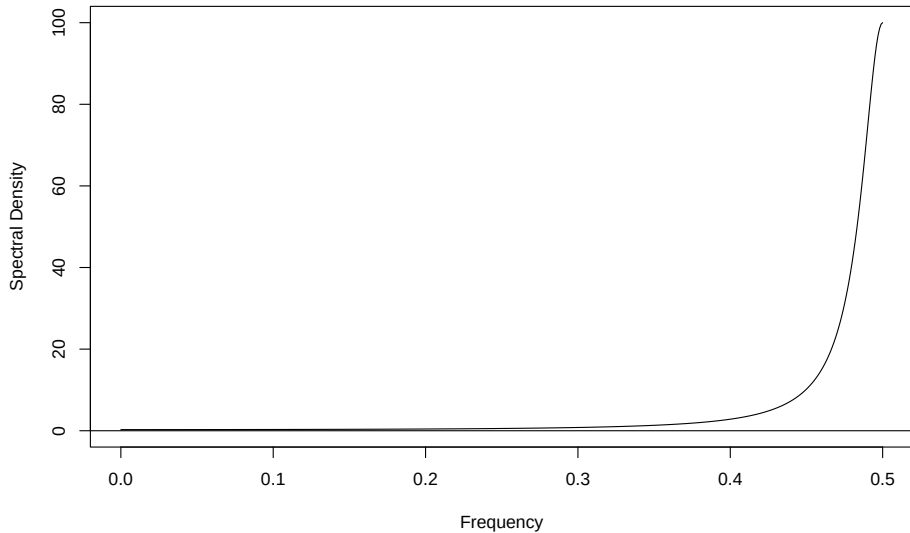
Example: $AR(1)$

- ▶ If $\phi_1 > 0$ (positive autocorrelation), spectrum is dominated by low frequency components—smooth in the time domain.
- ▶ If $\phi_1 < 0$ (negative autocorrelation), spectrum is dominated by high frequency components—rough in the time domain.

Example: $AR(1)$ Spectrum



Example: $AR(1)$ Spectrum



Example: $MA(1)$ Spectrum

$$X_t = W_t - \theta W_{t-1}.$$

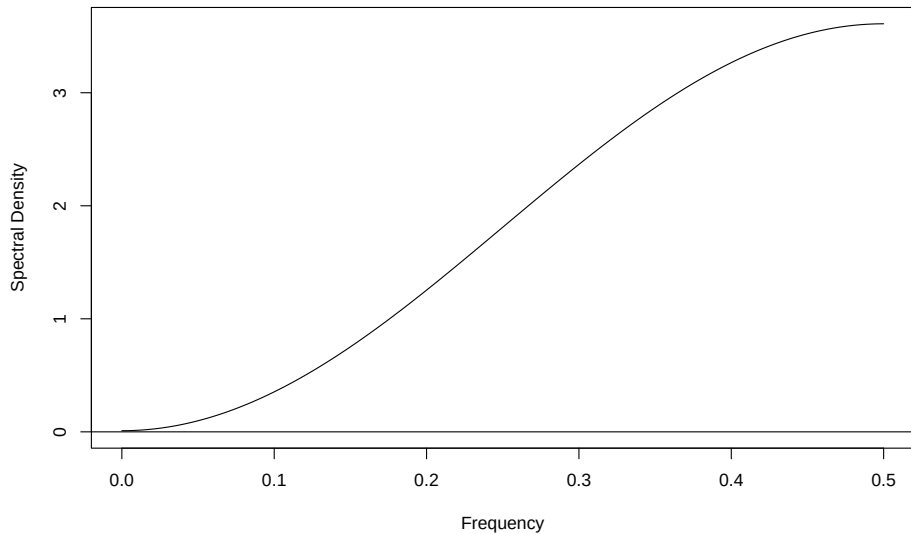
$$\gamma(h) = \begin{cases} \sigma_w^2(1 + \theta^2) & \text{if } h = 0, \\ -\sigma_w^2\theta & \text{if } |h| = 1, \\ 0 & \text{otherwise.} \end{cases}$$

$$\begin{aligned} S(\nu) &= \sum_{h=-1}^1 \gamma(h) e^{-2\pi i \nu h} \\ &= \gamma(0) + 2\gamma(1) \cos(2\pi\nu) \\ &= \sigma_w^2(1 + \theta^2 - 2\theta \cos(2\pi\nu)) \end{aligned}$$

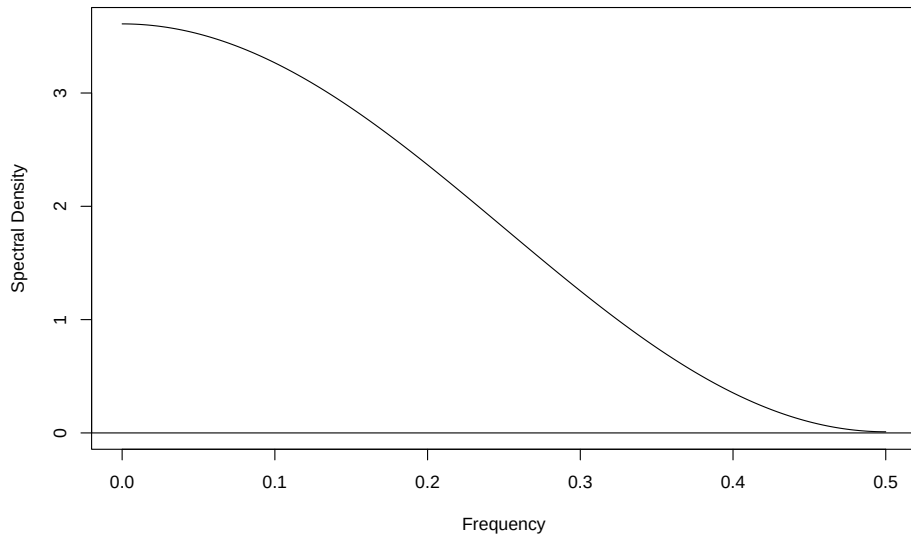
Example: $MA(1)$ Spectrum

- ▶ If $\theta < 0$ (positive autocorrelation), spectrum is dominated by low frequency components—smooth in the time domain.
- ▶ If $\theta > 0$ (negative autocorrelation), spectrum is dominated by high frequency components—rough in the time domain.

Example: $MA(1)$ Spectrum



Example: $MA(1)$ Spectrum



Discrete Spectral Distribution Function

- For $X_t = A \sin(2\pi\lambda t) + B \cos(2\pi\lambda t)$, we have $\gamma(h) = \sigma^2 \cos(2\pi\lambda h)$, and we can write

$$\gamma(h) = \int_{-1/2}^{1/2} e^{2\pi i\nu h} dF(\nu),$$

where F is the discrete distribution

$$F(\nu) = \begin{cases} 0 & \text{if } \nu < -\lambda, \\ \frac{\sigma^2}{2} & \text{if } -\lambda \leq \nu < \lambda, \\ \sigma^2 & \text{otherwise.} \end{cases}$$

The Spectral Distribution Function

- For any stationary $\{X_t\}$ with autocovariance γ , we can write

$$\gamma(h) = \int_{-1/2}^{1/2} e^{2\pi i\nu h} dF(\nu),$$

where F is the spectral distribution function of $\{X_t\}$.

- If γ is absolutely summable, F is continuous: $dF(\nu) = S(\nu)d\nu$, and we get

$$\gamma(h) = \int_{-1/2}^{1/2} e^{2\pi i\nu h} S(\nu) d\nu,$$

where $S(\nu)$ is the spectral density.

The Spectral Distribution Function

- ▶ For $X_t = \sum_{j=1}^k (A_j \sin(2\pi\nu_j t) + B_j \cos(2\pi\nu_j t))$, the spectral distribution function is $F(\nu) = \sum_{j=1}^k \sigma_j^2 F_j(\nu)$, where

$$F_j(\nu) = \begin{cases} 0 & \text{if } \nu < -\nu_j, \\ \frac{1}{2} & \text{if } -\nu_j \leq \nu < \nu_j, \\ 1 & \text{otherwise.} \end{cases}$$

- ▶ The heights of the jumps in the spectral distribution give the variances associated with the various periodic components, and the positions of the jumps indicate the frequencies of the periodic components.

The Spectral Distribution Function

For $0 \leq \nu_1 < \nu_2 \leq \frac{1}{2}$, the integral $\int_{\nu_1}^{\nu_2} dF(\nu)$, gives the portion of the (total) process variance $F\left(\frac{1}{2}\right) = \gamma_0$ that is attributable to frequencies in the range ν_1 to ν_2 .

LINEAR FILTERS

Linear Filters

- ▶ Linear filters are important in signal processing.
- ▶ They help in removing noise, extracting useful signals, and shaping frequency content (like low-pass and high-pass filters).
- ▶ Linear filters are also stable and fast, making them widely used in audio processing, communication systems, and medical signals.

Time-invariant linear filters

- ▶ A filter is an operator that maps a times series $\{X_t\}$ onto a times series $\{Y_t\}$.
- ▶ We can think of a general linear process $X_t = \sum_{j=0}^{\infty} \psi_j e_{t-j}$ as the output of a causal linear filter with a white noise input.
- ▶ A time series $\{Y_t\}$ is the output of a linear filter $A = \{a_{t,j} : t, j \in \mathbb{Z}\}$ with input $\{X_t\}$ if

$$Y_t = \sum_{j=-\infty}^{\infty} a_{t,j} X_j.$$

Time-invariant linear filters

- ▶ If $a_{t,t-j}$ is independent of t (i.e., $a_{t,t-j} = \psi_j$), then we say that the filter is **time-invariant**.
- ▶ If $\psi_j = 0$ for $j < 0$, we say the filter ψ is **causal**.
- ▶ We'll see that the name "filter" arises from the frequency domain viewpoint.

Time-Invariant Linear Filters: Examples

- ▶ $Y_t = X_{-t}$ is linear, but not time-invariant.
- ▶ $Y_t = \frac{1}{3}(X_{t-1} + X_t + X_{t+1})$ is linear, time-invariant, but not causal:

$$\psi_j = \begin{cases} \frac{1}{3} & \text{if } |j| \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- ▶ For polynomials $\phi(B)$, $\theta(B)$ with roots outside the unit circle, $\psi(B) = \frac{\theta(B)}{\phi(B)}$ is a linear, time-invariant, causal filter.

Frequency response of a time-invariant linear filter

- ▶ The operation $\sum_{j=-\infty}^{\infty} \psi_j X_{t-j}$ is called the **convolution** of X with ψ .
- ▶ The sequence ψ is also called the **impulse response**.
- ▶ Suppose that $\{X_t\}$ has spectral density $S_X(\nu)$ and ψ is stable, that is, $\sum_{j=-\infty}^{\infty} |\psi_j| < \infty$. Then

$$Y_t = \sum_{j=-\infty}^{\infty} \psi_j X_{t-j} = \psi(B)X_t$$

has spectral density

$$S_Y(\nu) = |\psi(e^{-2\pi i\nu})|^2 S_X(\nu).$$

Proof

$$\begin{aligned}\text{Cov}(Y_t, Y_{t-k}) &= \text{Cov}\left(\sum_{j=-\infty}^{\infty} \psi_j X_{t-j}, \sum_{s=-\infty}^{\infty} \psi_s X_{t-k-s}\right) \\&= \sum_{j=-\infty}^{\infty} \sum_{s=-\infty}^{\infty} \psi_j \psi_s \text{Cov}(X_{t-j}, X_{t-k-s}) \\&= \sum_{j=-\infty}^{\infty} \sum_{s=-\infty}^{\infty} \psi_j \psi_s \int_{-1/2}^{1/2} e^{2\pi i(s+k-j)f} S_X(f) df \\&= \int_{-1/2}^{1/2} \left(\sum_{j=-\infty}^{\infty} \psi_j e^{-2\pi i j f}\right) \left(\sum_{s=-\infty}^{\infty} \psi_s e^{2\pi i s f}\right) e^{2\pi i k f} S_X(f) df \\&= \int_{-1/2}^{1/2} \left|\sum_{s=-\infty}^{\infty} \psi_s e^{-2\pi i s f}\right|^2 e^{2\pi i k f} S_X(f) df \\&= \int_{-1/2}^{1/2} |\psi(e^{-2\pi i f})|^2 e^{2\pi i k f} S_X(f) df\end{aligned}$$

Proof

► So,

$$\text{Cov}(Y_t, Y_{t-k}) = \int_{-1/2}^{1/2} |\psi(e^{-2\pi if})|^2 S_X(f) e^{2\pi ifk} df$$

► But,

$$\text{Cov}(Y_t, Y_{t-k}) = \int_{-1/2}^{1/2} S_Y(f) e^{2\pi ifk} df$$

► Thus, we must have,

$$S_Y(f) = |\psi(e^{-2\pi if})|^2 S_X(f)$$

Frequency response of a time-invariant linear filter

- ▶ The function $\nu \mapsto \psi(e^{-2\pi i\nu})$ (the polynomial $\psi(z)$ evaluated on the unit circle) is known as the **frequency response or transfer function** of the linear filter.
- ▶ The squared modulus, $\nu \mapsto |\psi(e^{-2\pi i\nu})|^2$, is known as the **power transfer function** of the filter.

Linear Processes and Filters

- ▶ We have seen that a general linear process, $Y_t = \psi(B)e_t$, is a special case, since

$$S_Y(\nu) = |\psi(e^{-2\pi i\nu})|^2 S_e(\nu) = |\psi(e^{-2\pi i\nu})|^2 \sigma_e^2.$$

- ▶ When we pass a time series $\{X_t\}$ through a linear filter, the spectral density is multiplied, frequency-by-frequency, by the squared modulus of the frequency response $\nu \mapsto |\psi(e^{-2\pi i\nu})|^2$.
- ▶ This is a version of the equality $\text{Var}(aX) = a^2 \text{Var}(X)$, but the equality is true for the component of the variance at every frequency.
- ▶ This is also the origin of the name "filter".

Frequency Response: Examples

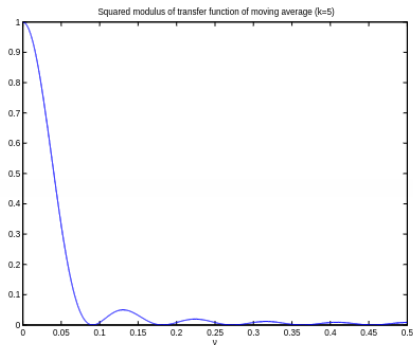
- Consider the moving average filter:

$$Y_t = \frac{1}{2k+1} \sum_{j=-k}^k X_{t-j}.$$

- This is a time-invariant linear filter (but not causal). Its transfer function is the Dirichlet kernel

$$\begin{aligned} \psi(e^{-2\pi i\nu}) &= D_k(2\pi\nu) = \frac{1}{2k+1} \sum_{j=-k}^k e^{-2\pi i j \nu} \\ &= \begin{cases} 1 & \text{if } \nu = 0, \\ \frac{\sin(2\pi(k+1/2)\nu)}{(2k+1)\sin(\pi\nu)} & \text{otherwise.} \end{cases} \end{aligned}$$

Example: Dirichlet Kernel



This is a **low-pass filter**: It preserves low frequencies and diminishes high frequencies. It is often used to estimate a monotonic trend component of a series.

Frequency Response: Examples (cont.)

- ▶ Consider the first difference

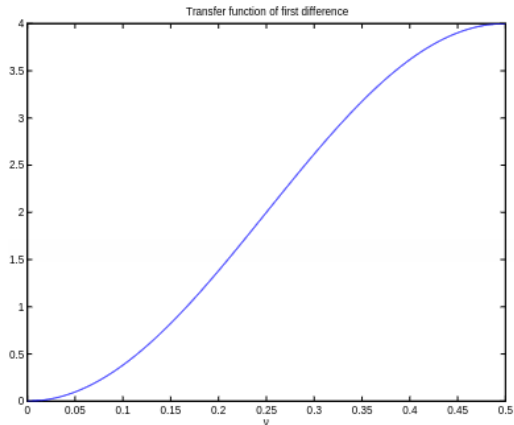
$$Y_t = (1 - B)X_t.$$

- ▶ This is a time-invariant, causal, linear filter. Its transfer function is

$$\psi(e^{-2\pi i\nu}) = 1 - e^{-2\pi i\nu},$$

$$\text{so } |\psi(e^{-2\pi i\nu})|^2 = 2(1 - \cos(2\pi\nu)).$$

Example: First Difference Kernel



This is a [high-pass filter](#): It preserves high frequencies and diminishes low frequencies. It is often used to eliminate a trend component of a series.

Spectral Density of a Linear Process

- ▶ If X_t is a linear process, it can be written $X_t = \sum_{i=0}^{\infty} \psi_i W_{t-i} = \psi(B) W_t$.
Then

$$S(\nu) = \sigma_w^2 |\psi(e^{-2\pi i \nu})|^2.$$

- ▶ That is, the spectral density $S(\nu)$ of a linear process measures the modulus of the ψ (MA(∞)) polynomial at the point $e^{2\pi i \nu}$ on the unit circle.

Spectral Density of a Linear Process

- For an $ARMA(p, q)$, $\psi(B) = \frac{\theta(B)}{\phi(B)}$, so

$$\begin{aligned} S(\nu) &= \sigma_w^2 \left| \frac{\theta(e^{-2\pi i\nu})}{\phi(e^{-2\pi i\nu})} \right|^2 \\ &= \sigma_w^2 \frac{\theta(e^{-2\pi i\nu})\theta(e^{2\pi i\nu})}{\phi(e^{-2\pi i\nu})\phi(e^{2\pi i\nu})} \\ &= \sigma_w^2 \frac{|1 - \theta_1 e^{-2\pi i\nu} - \dots - \theta_q e^{-2\pi i\nu q}|^2}{|1 - \phi_1 e^{-2\pi i\nu} - \dots - \phi_p e^{-2\pi i\nu p}|^2} \end{aligned}$$

- This is known as a rational spectrum.

Rational Spectra

- Consider the factorization of Θ and Φ as

$$\begin{aligned}\theta(z) &= \theta_q(z - z_1)(z - z_2) \cdots (z - z_q) \\ \phi(z) &= \phi_p(z - p_1)(z - p_2) \cdots (z - p_p),\end{aligned}$$

where $z_1, \dots, z_q$ and $p_1, \dots, p_p$ are called the zeros and poles.

$$S(\nu) = \sigma_w^2 \left| \frac{\theta_q \prod_{j=1}^q (e^{-2\pi i\nu} - z_j)}{\phi_p \prod_{j=1}^p (e^{-2\pi i\nu} - p_j)} \right|^2 = \sigma_w^2 \frac{\theta_q^2 \prod_{j=1}^q |e^{-2\pi i\nu} - z_j|^2}{\phi_p^2 \prod_{j=1}^p |e^{-2\pi i\nu} - p_j|^2}.$$

- As ν varies from 0 to $1/2$, $e^{-2\pi i\nu}$ moves clockwise around the unit circle from 1 to $e^{-\pi i} = -1$.
- And the value of $S(\nu)$ goes up as this point moves closer to (further from) the poles p_j (zeros z_j).

Rational Spectra: Examples

- ▶ Recall $AR(1)$: $\phi(z) = 1 - \phi_1 z$. The pole is at $1/\phi_1$.
 - If $\phi_1 > 0$, the pole is to the right of 1, so the spectral density decreases as ν moves away from 0.
 - If $\phi_1 < 0$, the pole is to the left of -1, so the spectral density is at its maximum when $\nu = 0.5$.
- ▶ Recall $MA(1)$: $\theta(z) = 1 + \theta_1 z$. The zero is at $-1/\theta_1$.
 - If $\theta_1 > 0$, the zero is to the left of -1, so the spectral density decreases as ν moves towards -1.
 - If $\theta_1 < 0$, the zero is to the right of 1, so the spectral density is at its minimum when $\nu = 0$.

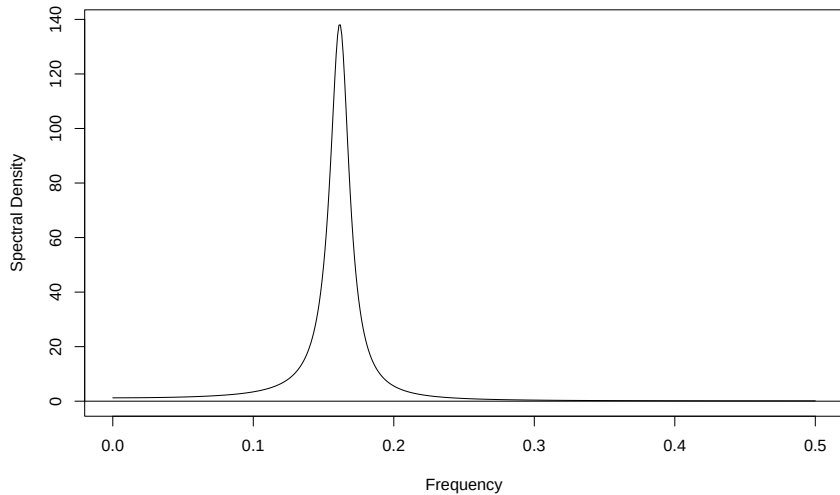
Example: $AR(2)$

- ▶ Consider $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + e_t$ with $\phi_1 = 1$, $\phi_2 = -0.9$, and $\sigma_e^2 = 1$.
- ▶ In this case, the poles are at
 $p_1, p_2 \approx 0.5555 \pm i0.8958 \approx 1.054e^{\pm i1.01567} \approx 1.054e^{\pm 2\pi i0.16165}$.
- ▶ Thus, we have

$$S(\nu) = \frac{\sigma_w^2}{\phi_2^2 |e^{-2\pi i\nu} - p_1|^2 |e^{-2\pi i\nu} - p_2|^2},$$

and this gets very peaked when $e^{-2\pi i\nu}$ passes near $1.054e^{-2\pi i0.16165}$.

Example: $AR(2)$ Spectrum



ESTIMATING THE SPECTRAL DENSITY

Estimating the Spectrum: Non-parametric Approaches

- ▶ We have seen that the spectral density gives an alternative view of stationary time series.
- ▶ Given a realization $x_1, \dots, x_n$ of a time series, how can we estimate the spectral density?
 - One approach: replace $\gamma(\cdot)$ in the definition, with the sample autocovariance $\hat{\gamma}(\cdot)$:

$$S(\nu) = \sum_{h=-\infty}^{\infty} \gamma(h) e^{-2\pi i \nu h},$$

- Another approach, called the periodogram: compute $I(\nu)$, the squared modulus of the discrete Fourier transform (at frequencies $\nu = k/n$).

Estimating the Spectrum: Sample Autocovariance

- **Idea:** use the sample autocovariance $\hat{\gamma}(\cdot)$, defined by

$$\hat{\gamma}(h) = \frac{1}{n} \sum_{t=1}^{n-|h|} (x_{t+|h|} - \bar{x})(x_t - \bar{x}), \text{ for } -n < h < n,$$

as an estimate of the autocovariance $\gamma(\cdot)$, and then use a sample version of

$$S(\nu) = \sum_{h=-\infty}^{\infty} \gamma(h) e^{-2\pi i \nu h},$$

- That is, for $-1/2 \leq \nu \leq 1/2$, estimate $f(\nu)$ with

$$\hat{S}(\nu) = \sum_{h=-n+1}^{n-1} \hat{\gamma}(h) e^{-2\pi i \nu h}.$$

Estimating the Spectrum: Periodogram

- **Discrete Fourier Transform:** For a sequence $(x_1, \dots, x_n)$, define the discrete Fourier transform (DFT) as $(X(\nu_0), X(\nu_1), \dots, X(\nu_{n-1}))$, where

$$X(\nu_k) = \frac{1}{\sqrt{n}} \sum_{t=1}^n x_t e^{-2\pi i \nu_k t},$$

and $\nu_k = k/n$ (for $k = 0, 1, \dots, n-1$) are called the Fourier frequencies.

- First, let's show that we can view the DFT as a representation of x in a different basis, the Fourier basis.

Periodogram

- ▶ Consider the following set of n orthonormal vectors in $\mathbb{C}^n$:

$$\left\{ e_j = \frac{1}{\sqrt{n}} \left(e^{2\pi i \nu_j}, e^{2\pi i \cdot 2\nu_j}, \dots, e^{2\pi i \cdot n\nu_j} \right)' : j = 0, \dots, n-1 \right\}.$$

- ▶ So we can represent the real vector $x = (x_1, \dots, x_n)' \in \mathbb{C}^n$ in terms of this orthonormal basis,

$$x = \sum_{j=0}^{n-1} \langle e_j, x \rangle e_j = \sum_{j=0}^{n-1} X(\nu_j) e_j.$$

- ▶ That is, the vector of discrete Fourier transform coefficients $(X(\nu_0), \dots, X(\nu_{n-1}))$ is the representation of x in the Fourier basis.

Periodogram

- ▶ The periodogram is defined as

$$I(\nu_j) = |X(\nu_j)|^2 = \frac{1}{n} \left| \sum_{t=1}^n e^{-2\pi i t \nu_j} x_t \right|^2 = X_c^2(\nu_j) + X_s^2(\nu_j),$$

where

$$X_c(\nu_j) = \frac{1}{\sqrt{n}} \sum_{t=1}^n \cos(2\pi t \nu_j) x_t,$$

$$X_s(\nu_j) = \frac{1}{\sqrt{n}} \sum_{t=1}^n \sin(2\pi t \nu_j) x_t.$$

Periodogram

- ▶ Since $I(\nu_j) = |X(\nu_j)|^2$ for one of the Fourier frequencies $\nu_j = j/n$ (for $j = 0, 1, \dots, n-1$), the orthonormality of the e_j implies that we can write

$$x^*x = \left(\sum_{j=0}^{n-1} X(\nu_j) e_j \right)^* \left(\sum_{j=0}^{n-1} X(\nu_j) e_j \right) = \sum_{j=0}^{n-1} |X(\nu_j)|^2 = \sum_{j=0}^{n-1} I(\nu_j).$$

- ▶ For $\bar{x} = 0$, we can write this as

$$\hat{\sigma}_x^2 = \frac{1}{n} \sum_{t=1}^n x_t^2 = \frac{1}{n} \sum_{j=0}^{n-1} I(\nu_j).$$

Estimating the Spectrum: Periodogram

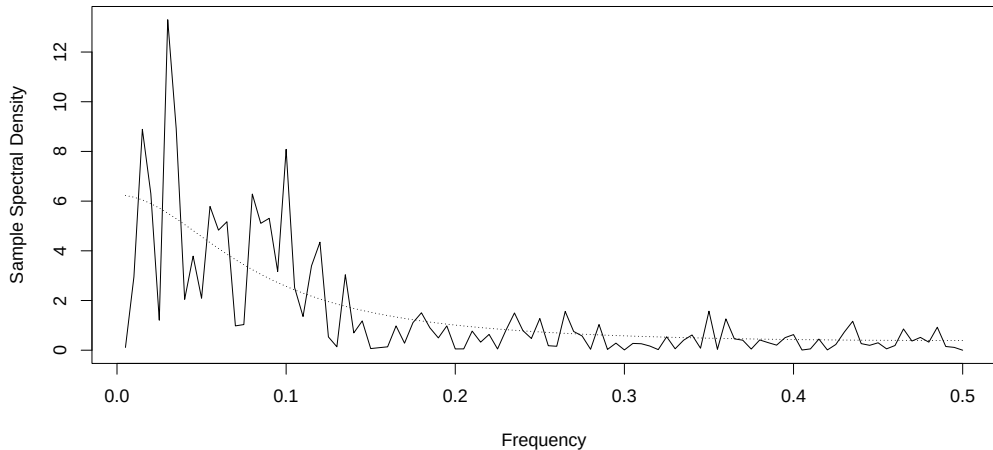
- ▶ At a Fourier frequency $\nu = \nu_j$, the periodogram is equivalent to evaluating the spectral density $\hat{S}(\nu)$ using the sample autocovariance function.
- ▶ The two are exactly the same for all Fourier frequencies ν_j with $j \in 1, \dots, n-1$, or if the sample mean $\bar{x} = 0$. However, at the zero frequency $\nu_0 = 0$, the two differ if $\bar{x} \neq 0$.

Estimating the Spectrum: Periodogram

$$\begin{aligned} I(\nu_j) &= \frac{1}{n} \left| \sum_{t=1}^n e^{-2\pi i t \nu_j} x_t \right|^2 \\ &= \frac{1}{n} \left| \sum_{t=1}^n e^{-2\pi i t \nu_j} (x_t - \bar{x}) \right|^2 \quad \text{for } \nu_j = \frac{j}{n}, j = 1, \dots, n-1 \\ &= \frac{1}{n} \left(\sum_{t=1}^n e^{-2\pi i t \nu_j} (x_t - \bar{x}) \right) \left(\sum_{t=1}^n e^{2\pi i t \nu_j} (x_t - \bar{x}) \right) \\ &= \frac{1}{n} \sum_{s,t} e^{-2\pi i (s-t) \nu_j} (x_t - \bar{x})(x_s - \bar{x}) \\ &= \sum_{h=-n+1}^{n-1} \hat{\gamma}(h) e^{-2\pi i h \nu_j} \quad (h = t - s) \end{aligned}$$

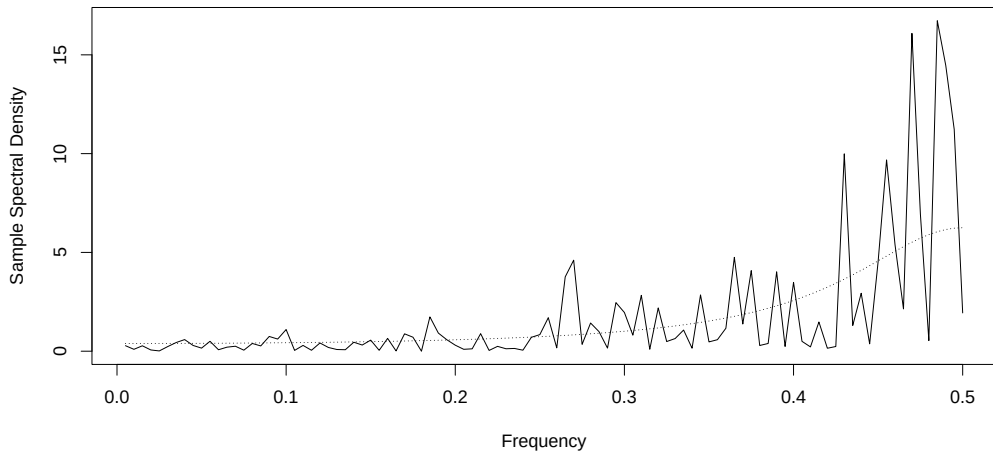
Example: $AR(1)$ Periodogram with $\phi = 0.6$

Spectral Density: Sample (Raw Periodogram) vs Theoretical (Dotted)



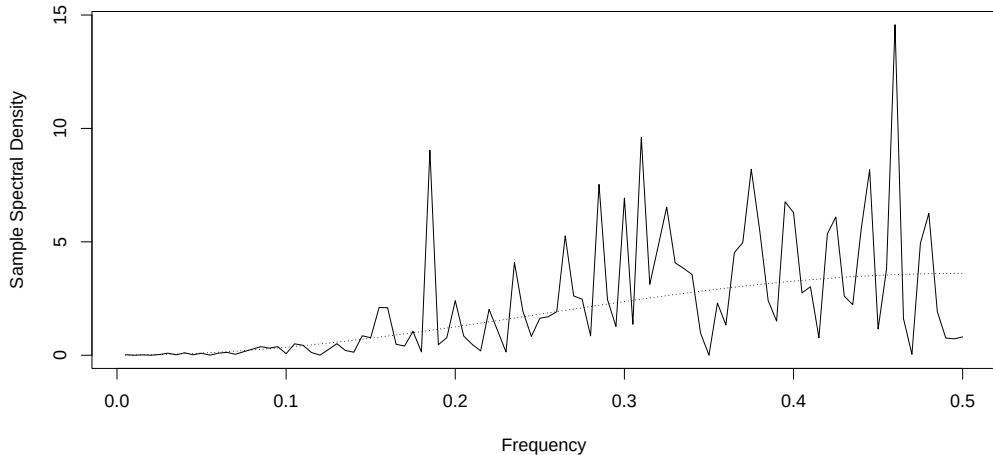
Example: $AR(1)$ Periodogram with $\phi = -0.6$

Spectral Density: Sample (Raw Periodogram) vs Theoretical (Dotted)



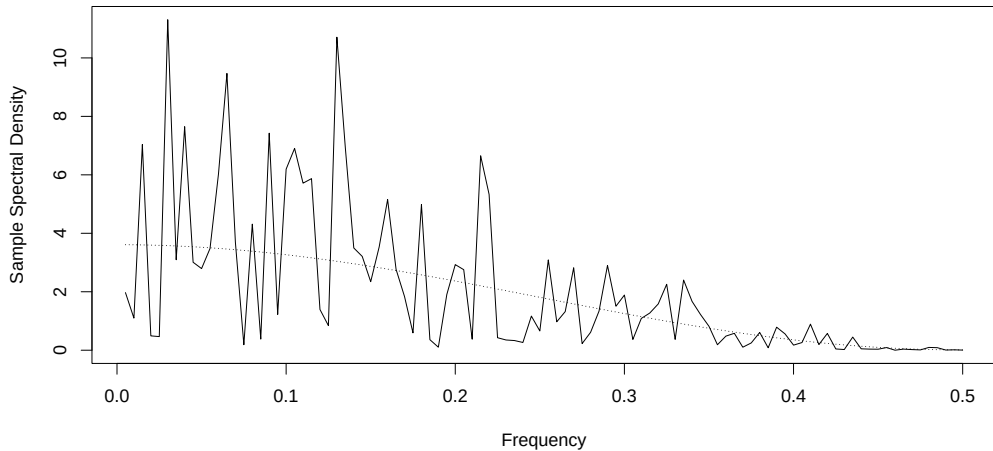
Example: $MA(1)$ Periodogram with $\theta = 0.9$

Spectral Density: Sample (Raw Periodogram) vs Theoretical (Dotted)



Example: $MA(1)$ Periodogram with $\theta = -0.9$

Spectral Density: Sample (Raw Periodogram) vs Theoretical (Dotted)



Asymptotic Properties of the Periodogram

- ▶ **Gaussian White Noise:** If $X_1, \dots, X_n$ are i.i.d. $\mathcal{N}(0, \sigma^2)$ (Gaussian white noise; $f(\nu) = \sigma^2$), then the $X_c(\nu_j)$ and $X_s(\nu_j)$ are all i.i.d. $\mathcal{N}(0, \sigma^2/2)$. Thus,

$$\frac{2}{\sigma^2} I(\nu_j) = \frac{2}{\sigma^2} (X_c^2(\nu_j) + X_s^2(\nu_j)) \sim \chi_2^2$$

- ▶ So for the case of Gaussian white noise, the periodogram has a chi-squared distribution that depends on the variance σ^2 (which, in this case, is the spectral density).

General Asymptotic Properties

- ▶ Under more general conditions (e.g., normal $\{X_t\}$, or linear process $\{X_t\}$ with rapidly decaying ACF), the $X_c(\nu_j)$, $X_s(\nu_j)$ are all asymptotically independent and $N(0, S(\nu_j)/2)$.
- ▶ Consider a frequency ν . For a given value of n , let $\hat{\nu}(n)$ be the closest Fourier frequency (that is, $\hat{\nu}(n) = j/n$ for a value of j that minimizes $|\nu - j/n|$). As n increases, $\hat{\nu}(n) \rightarrow \nu$, and (under the same conditions that ensure the asymptotic normality and independence of the sine/cosine transforms), $S(\hat{\nu}(n)) \rightarrow S(\nu)$.

General Asymptotic Properties

- In this case, we have:

$$\frac{2}{S(\nu)} I(\hat{\nu}(n)) = \frac{2}{S(\nu)} (X_c^2(\hat{\nu}(n)) + X_s^2(\hat{\nu}(n))) \xrightarrow{d} \chi_2^2$$

- Thus,

$$E(I(\hat{\nu}(n))) \rightarrow S(\nu),$$

but

$$\text{Var}(I(\hat{\nu}(n))) \rightarrow S(\nu)^2.$$

Estimating the Spectrum: Discussion

- ▶ The two estimation approaches are identical at the Fourier frequencies $\nu = k/n$.
- ▶ The asymptotic expectation of the periodogram $I(\nu)$ is $S(\nu)$. We can derive some asymptotic properties, and hence do hypothesis testing.
- ▶ Unfortunately, the asymptotic variance of $I(\nu)$ is **constant**. Thus, the periodogram is **not a consistent estimator** of $S(\nu)$.
- ▶ We can reduce the variance by smoothing the periodogram - (averaging over adjacent frequencies). If we average over a narrower range as $n \rightarrow \infty$, we can obtain a consistent estimator of the spectral density.

Smoothed Periodogram

Averaging the Periodogram

If we assume $S(\nu)$ is approximately constant in a band of frequencies $\left[\nu_k - \frac{L}{2n}, \nu_k + \frac{L}{2n}\right]$, we can average the periodogram over this band:

$$\begin{aligned}\hat{S}(\nu_k) &= \frac{1}{L} \sum_{l=-\frac{L-1}{2}}^{\frac{L-1}{2}} I\left(\nu_k - \frac{l}{n}\right) \\ &= \frac{1}{L} \sum_{l=-\frac{L-1}{2}}^{\frac{L-1}{2}} \left(X_c^2\left(\nu_k - \frac{l}{n}\right) + X_s^2\left(\nu_k - \frac{l}{n}\right) \right)\end{aligned}$$

Smoothed Periodogram

Asymptotic Properties

Under general conditions, the $X_c\left(\nu_k - \frac{l}{n}\right)$ and $X_s\left(\nu_k - \frac{l}{n}\right)$ are asymptotically independent and $N\left(0, f\left(\nu_k - \frac{l}{n}\right)/2\right)$. Thus,

$$\begin{aligned}\mathbb{E}(\hat{S}(\hat{\nu}(n))) &\rightarrow S(\nu) \\ \text{Var}(\hat{S}(\hat{\nu}(n))) &\rightarrow \frac{S^2(\nu)}{L}\end{aligned}$$

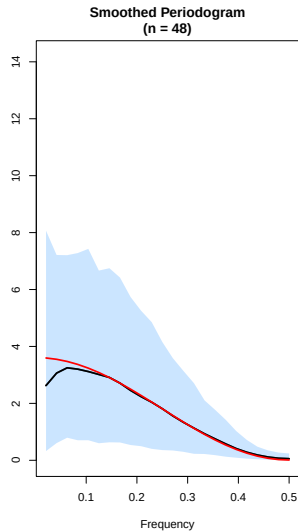
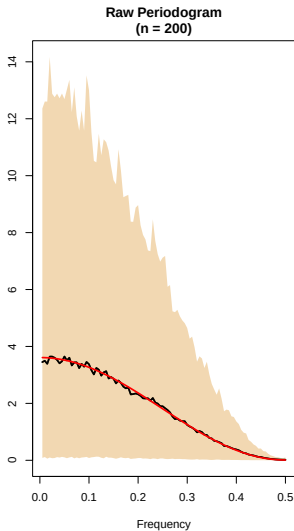
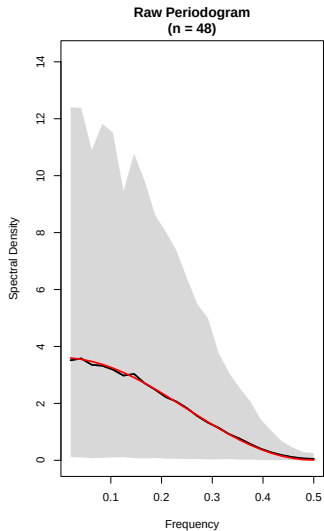
Bias-Variance Trade-off

1. Our assumption that S is approximately constant on $\left[\nu - \frac{L}{2n}, \nu + \frac{L}{2n}\right]$ becomes worse as L increases, so the difference between $\hat{S}(\hat{\nu}(n))$ and $S(\nu)$ (the bias) will increase with L .
2. The variance of our estimate, $\text{Var}\left(\hat{S}(\hat{\nu}(n))\right)$ decreases with L .

Spectral Density Estimation: Simulation Study

- ▶ We simulated 1000 replications of an $MA(1)$ process with parameter $\theta = -0.9$. The white noise input was drawn from a t -distribution with 5 degrees of freedom, scaled to have unit variance.
- ▶ For each replication, we computed:
 - The raw periodogram for $n = 48$.
 - The raw periodogram for $n = 200$.
 - The smoothed periodogram for $n = 48$.
- ▶ The plots show:
 - The average estimated spectral densities.
 - 95% confidence intervals across replications.
 - The true spectral density (in red).

Spectral Density Estimation: Simulation Study



Spectral Density Estimation: Simulation Study

► Key observations:

- The periodogram is an approximately unbiased estimator.
- Increasing sample size does not reduce variance.
- Smoothing reduces variance.

Parametric vs Nonparametric Estimation

- ▶ **Parametric estimation:** Estimate a model with a fixed number of parameters.
- ▶ **Nonparametric estimation:** Model is specified by a number of parameters that can grow with the sample size.
- ▶ Smoothed periodogram estimates are **nonparametric**:
 - Spectral density is estimated at each Fourier frequency.
 - Number of frequencies grows with sample size.
- ▶ Time domain models like linear processes are **parametric**:
 - An $ARMA(p, q)$ process is fully described by $p + q + 1$ parameters.

ARMA Spectral Estimation

- ▶ *ARMA* models can be used:
 - Estimate *ARMA*(p, q) parameters.
 - Compute:

$$\hat{f}(\nu) = \hat{\sigma}_w^2 \left| \frac{\hat{\theta}(e^{-2\pi i\nu})}{\hat{\phi}(e^{-2\pi i\nu})} \right|^2$$

Parametric vs Nonparametric Estimation

► Parametric advantages:

- Better frequency resolution for few peaks.
- Lower variance with fewer parameters.
- Even $p = 2$ can allow sharp peak at one frequency.

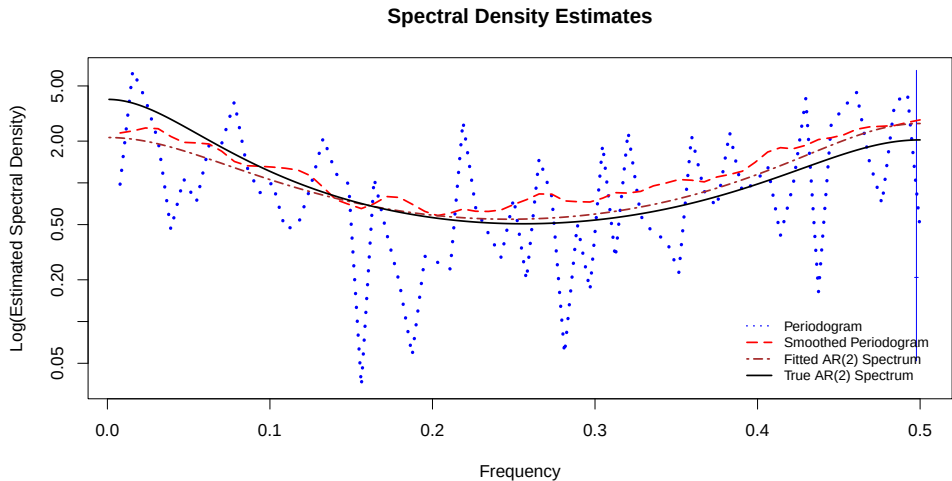
► Nonparametric trade-off:

- To reduce variance, bandwidth must not be too small.
- This limits frequency resolution.
- Important when multiple nearby peaks exist.

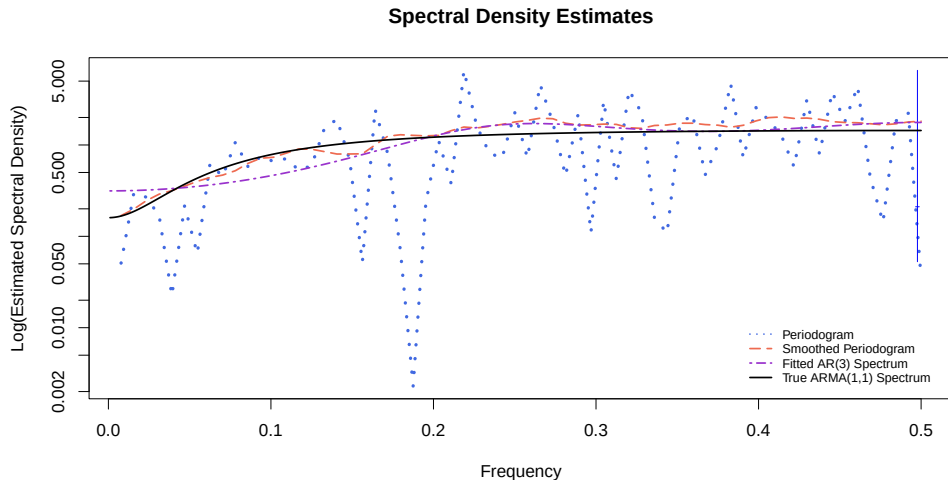
► Parametric disadvantage:

- Inflexibility due to restrictive *ARMA* model class.

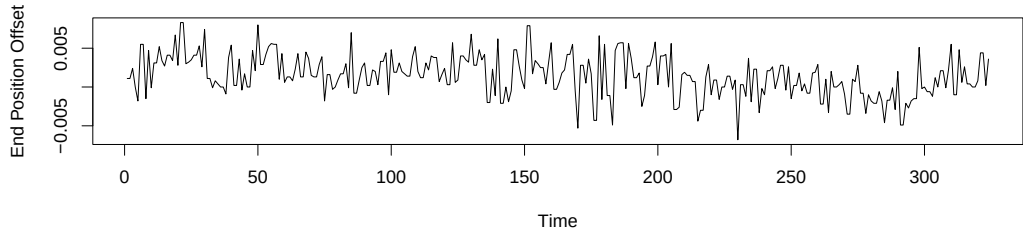
$AR(2)$ with $\phi_1 = 0.1, \phi_2 = 0.4$



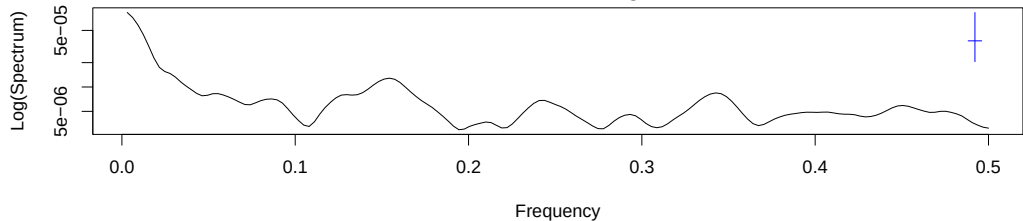
$ARMA(1, 1)$ with $\phi = 0.5, \theta = 0.8$



Robot Arm Positioning



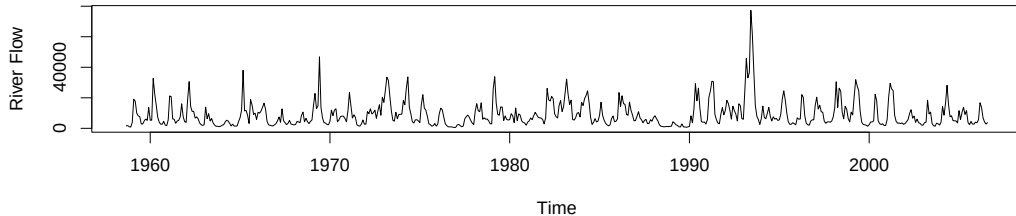
Series: x
Smoothed Periodogram



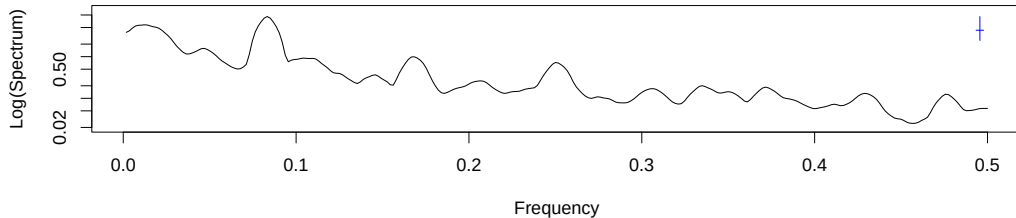
Robot Arm Positioning: Spectral Analysis

- ▶ An industrial robot was put through a sequence of maneuvers.
- ▶ The distance from a desired target end position was recorded (in inches).
- ▶ This process was repeated 324 times, forming a time series of length 324.
- ▶ Spectral analysis reveals:
 - A prominent peak at a frequency around 0.15, appearing in both the raw and smoothed estimates.
 - This peak is likely to be a real feature of the data.
 - Additional peaks at higher frequencies may be **spurious**.
- ▶ A large amount of spectral power is observed at **very low frequencies**, which is consistent with the **slowly drifting behavior** visible in the time series.

Iowa River Flow



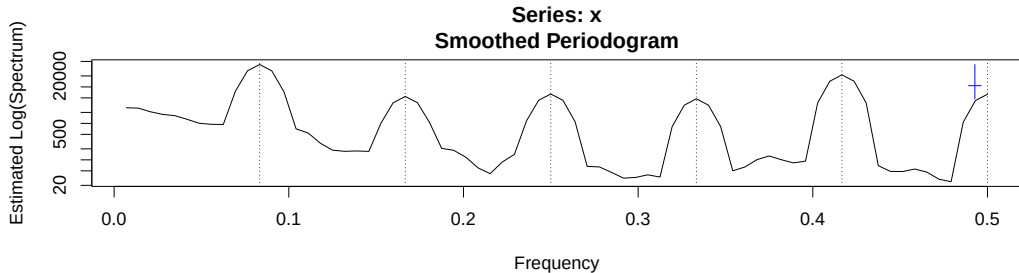
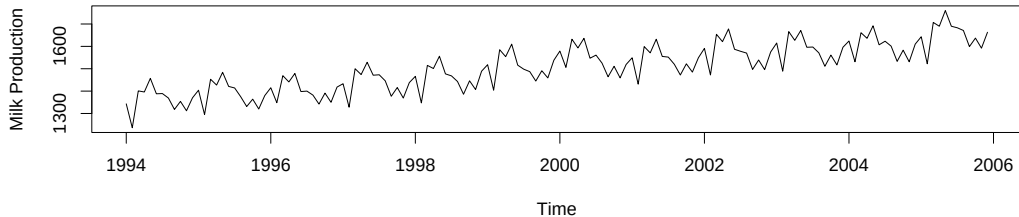
Series: x
Smoothed Periodogram



Iowa River Flow: Spectral Analysis

- ▶ Monthly river flow data for the Iowa River at Wapello, Iowa.
- ▶ Time period: September 1958 through August 2006.
- ▶ Sample size: 576 observations (24 years of monthly data).
- ▶ The original data are highly **skewed**, but this was greatly improved by applying a **log transformation**.
- ▶ Spectral analysis highlights:
 - A **prominent peak** at frequency $1/12$, indicating strong **annual seasonality**.
 - **Secondary peaks** at $f \approx 0.17$ and $f \approx 0.25$.
 - These correspond to **higher harmonics** of the fundamental annual frequency.

U.S. Monthly Milk Production



U.S. Monthly Milk Production: Spectral Analysis

- ▶ Data: U.S. monthly milk production from January 1994 through December 2005.
- ▶ The series shows:
 - A **substantial upward trend**.
 - Clear **seasonal patterns**.
- ▶ To analyze the seasonality:
 - We first remove the upward trend using a **simple linear regression**.
 - The residuals from the trend model represent the **detrended seasonal component**.
- ▶ Spectral analysis of the residuals shows:
 - Peaks at frequencies $\frac{1}{12}, \frac{2}{12}, \dots, \frac{6}{12}$.
 - The peak at $\frac{1}{12}$ has the **highest power**, indicating strong **annual seasonality**.